

SA-521-500

|                                                                        |                          |                                    |                            |
|------------------------------------------------------------------------|--------------------------|------------------------------------|----------------------------|
| <b>VARADA ENERGIES PVT LTD. ( AN ISO 9001-2015 CERTIFIED COMPANY )</b> |                          |                                    |                            |
| <b>STANDARD- BS 5514</b>                                               | <b>ISO- 8528</b>         |                                    |                            |
| <b>GENERAL INFORMATION</b>                                             |                          |                                    |                            |
| <b>Customer Name</b>                                                   |                          | <b>Customer Order No.</b>          |                            |
|                                                                        |                          | <b>Test Cell No.</b>               | 1                          |
| <b>PRODUCT INFORMATION</b>                                             |                          |                                    |                            |
| <b>Model</b>                                                           | VEPL-500-CDP             | <b>Genset Serial No.</b>           | VEPL/2021-22/0008          |
| <b>Engine Serial No.</b>                                               | MGBF 5083N14201F         | <b>Alternator Serial No.</b>       | N21C 115831                |
| <b>Brand Name</b>                                                      | PERKINS                  | <b>Alternator Make.</b>            | STAMFORD                   |
| <b>Fuel Type</b>                                                       | HSD                      | <b>Controller Type</b>             | DEEP SEA                   |
| <b>Governor Type</b>                                                   | ELECTRONIC               | <b>Cooling System Type</b>         | WATER COOLED               |
| <b>TEST CONDITION</b>                                                  |                          |                                    |                            |
| <b>Altitude (Meters ASL)</b>                                           | 1000                     | <b>Oil Specifications</b>          | PERKINS MAKE               |
| <b>Ambient Temp. (C)</b>                                               | 33                       | <b>Humidity (%)</b>                | 41                         |
| <b>Air Filter Inlet Temp (C)</b>                                       | 38                       | <b>Coolant Mix</b>                 | PERKINS MAKE               |
| <b>PRODUCT OPTIONS CHECKED</b>                                         |                          |                                    |                            |
| (e) Emergency Stop                                                     | (e) Panel Hooter         | (e) Low Coolant Warning            | (o) High Fuel S/Down       |
| (e) Over Load Protection                                               | (e) High Water Temp      | (o) LLET Warning                   | (o) High Fuel Warning      |
| (e) Over Speed Protection                                              | (e) Low Lub Oil Pressure | (o) Low Fuel S/Down                | (e) Parallel Bus PT Module |
| (e) Under Voltage Protection                                           | (e) Low Coolant S/Down   | (e) Low Fuel Warning               | (o) Earth Fault S/Down     |
| <b>TECHNICAL DATA (Declared)</b>                                       |                          |                                    |                            |
| <b>Service Rating</b>                                                  | 500 kVA                  | <b>Power Factor</b>                | 0.8                        |
| <b>Power Output (kW)</b>                                               | 400                      | <b>Rated Output (kW)</b>           | 400                        |
| <b>Rated Voltage (V)</b>                                               | 400                      | <b>Rated Current (A)</b>           | 590                        |
| <b>Frequency (Hz)</b>                                                  | 60                       | <b>Speed (RPM)</b>                 | 1800                       |
| <b>Number of Phases</b>                                                | 3                        | <b>Performance Class</b>           | H                          |
| <b>TECHNICAL DATA (Measured)</b>                                       |                          |                                    |                            |
| <b>Results (No Load)</b>                                               |                          |                                    |                            |
| <b>Voltage</b>                                                         | 1 - 2 : 402 Volts        | 2 - 3 : 401 Volts                  | 1 - 3 : 403 Volts          |
| <b>No Load Frequency</b>                                               | 60                       | <b>No Load Coolant Temp. (C)</b>   | 71                         |
| <b>No Load Lube Press</b>                                              | 3.90 Bar                 |                                    |                            |
| <b>Result (Full Load)</b>                                              |                          |                                    |                            |
| <b>Voltage</b>                                                         | 1 - 2 : 405 Volts        | 2 - 3 : 404 Volts                  | 1 - 3 : 400 Volts          |
| <b>Current</b>                                                         | 1 : 590 Amps             | 2 : 585 Amps                       | 3 : 590 Amps               |
| <b>Full Load Frequency</b>                                             | 60 Hz.                   | <b>Full Load Coolant Temp. (C)</b> | 88                         |
| <b>Full Load Lube Press</b>                                            | 3.78 Bar                 | <b>Power Factor</b>                | 1.0                        |
| <b>Prime set tested to 110% of Full Load (if required)</b>             |                          |                                    |                            |
|                                                                        |                          | <b>Date</b>                        | 15.05.2021                 |
|                                                                        |                          | <b>Run Hours</b>                   | 45 mins                    |
| <b>TYPE TEST REPORT REF NO</b>                                         | CE-JOB-MUM-20-000073-001 |                                    |                            |



## Engine Test Certificate

Perkins Engines Company Limited  
Eastfield, Peterborough PE1 5FQ  
Great Britain

SG-521-500



Aug 14, 2020

This engine has been tested to its maximum rating in accordance with Perkins defined requirements and has performed satisfactorily.

The electronics on this engine have been preset and when the engine has been run-in, the gross power output will be within +/-3% of the gross rating listed below, provided the exhaust and induction systems; fuel and lubricating oils are in accordance with Perkins defined requirements.

| Engine Type   | Perkins Serial Number | Customer |
|---------------|-----------------------|----------|
| 2506A-E15TAG2 | MGBF5083N14201F       | Y932     |

| Reference                        | Technical Data      |
|----------------------------------|---------------------|
| Emission Capability              | UNREGULATED Low SFC |
| Engine Speed                     | 50 Hz @1500RPM      |
| Engine Type                      | 2506A-E15TAG2       |
| Fuel Consumption - Prime Power   | 192 g/kWH           |
| Fuel Consumption - Standby Max   | 193 g/kWH           |
| Gross Engine Power - Prime Power | 458 kW              |
| Gross Engine Power - Standby Max | 503 kW              |
| Gross Rating                     | 458 kW              |
| Net Rating                       | 435 kW              |

This engine was built and tested in accordance with our approved ISO 9001 Quality Procedure and stated Rating, as determined by the ISO 3046 (BS5541) Standard.

Performance certified as satisfactory.

Signed,

For and on behalf of Perkins Engines Company Limited - Large Engines Division.

**MUST BE KEPT WITH ENGINE AT ALL TIMES**